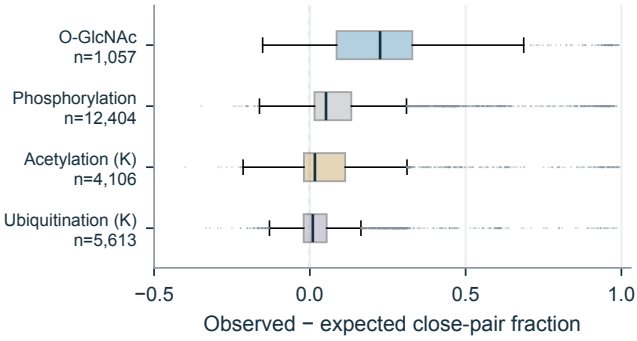
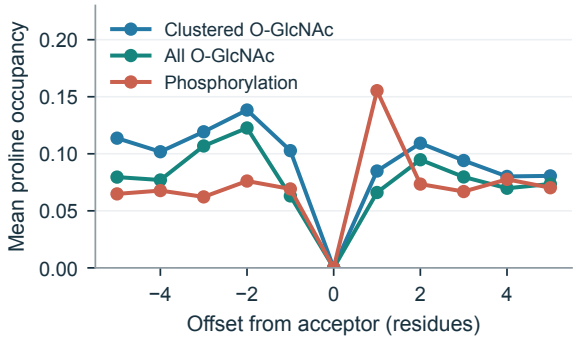
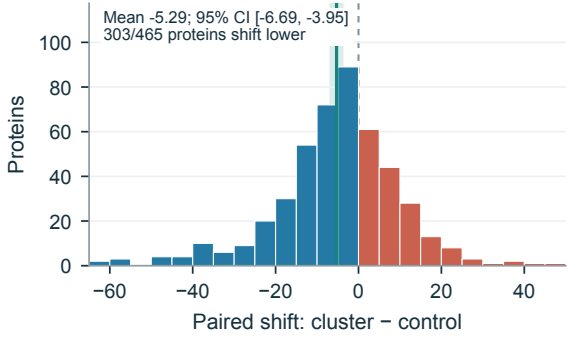
A Protein-level clustering distributions



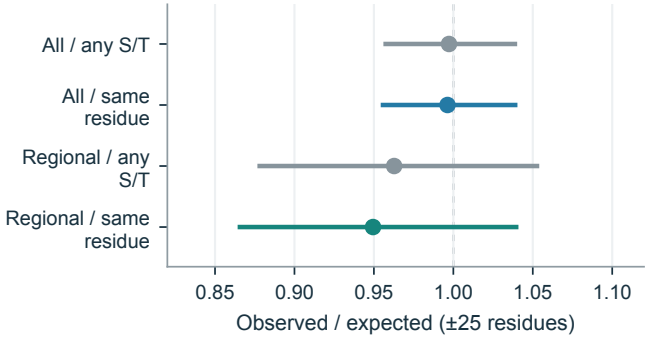
B Proline around the acceptor



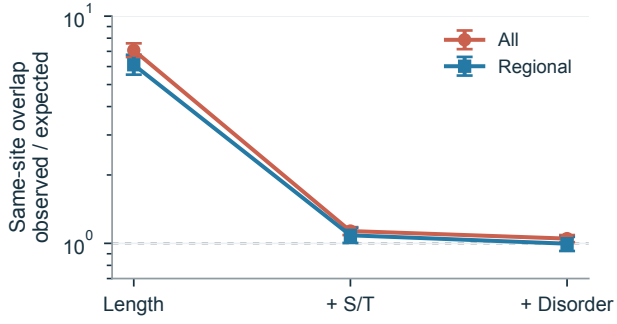
C Paired kinase shifts across proteins



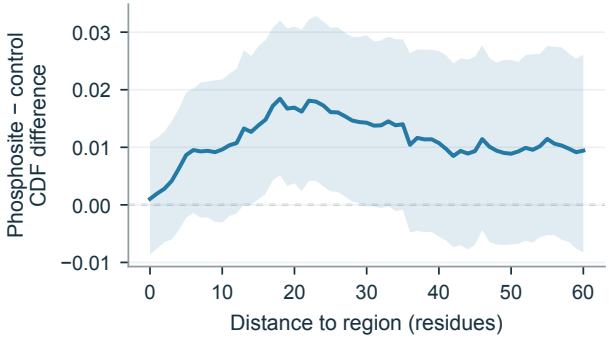
D Same-site overlap: primary local control



E Progressively conditioned nulls



F Distance after covariate matching



A: Protein-level observed-minus-expected close-pair fractions at radius 10; boxes show median/IQR, whiskers 1.5 IQR, dots outliers. Cohorts differ; K-based modifications use different eligible residues. C: All 465 paired protein shifts in high-scoring kinase count; teal marks archived mean and 95% CI. D: Primary local controls; all 12 remain in S2. Static co-occurrence does not establish competition.